

Construction of Aamli Fala Bridge and Road

Widening of (RCC Slab Culvert)

10 M.-

3 SPAN

DETAILS

S.No.	Description of Item.	No.	L	W	H/D	Quantity
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- 1 Earthwork in excavation for foundation of structure complete with all lift, dewatering, shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MoRTH Specification : Clause 304]. In ordinary soil [MoRTH Specification : clause 301] BY MECHANICAL MEANS Depth upto 3m

Abutment	2	8.50	8.80	1.20	=	179.52	cum
Piers	2	5.90	13.10	1.20	=	185.50	cum
For Wall	1	60.00	3.50	1.20	=	252.00	cum
Total					=	617.02	cum

- 2 Earthwork in excavation for foundation of structure complete with all lift, dewatering, shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MoRTH Specification : Clause 304]. In Ordinary/Soft rock (not requiring blasting) [MoRTH Specification : Clause 301] BY MECHANICAL MEANS

Abutment	2	8.50	8.80	1.50	=	224.40	cum
Piers	2	5.90	13.10	1.50	=	231.87	cum
For Wall	1	60.00	3.50	1.00	=	210.00	cum
Total					=	666.27	cum

- 3 Earthwork in excavation for foundation of structure complete with all lift, dewatering, shoring and shuttering etc. including refilling of trenches in 250 mm layer, ramming, watering and disposal of surplus soil within a lead of 1000 meters, as per drawing and technical Earth in excavation for foundation of specification. [MoRTH Specification : Clause 304]. In Hard Rock [MoRTH Specification : Clause 301, 302, 303]. Blasting Prohibited

Abutment	2	8.50	8.80	1.50	=	224.40	cum
Piers	2	5.90	13.10	1.50	=	231.87	cum
Total					=	456.27	cum

S.No.	Description of Item.	No.	L	W	H/D	Quantity
4	Structural cement concrete (Design Mix) B-2 In open foundations, with form work, at any level. Cement concrete for plain/reinforced concrete in open foundations as per drawing and technical specification. [MoRTH Specification : Section 1700 & 1500]. With 20mm nominal size aggregate P.C.C. M-20 grade	2 2 1	8.50 5.90 60.00	8.80 13.10 3.20	0.15 0.15 0.15	= = = Total = 22.44 cum 23.19 cum 28.80 cum 74.43 cum
5	Structural cement concrete (Design Mix) B-2 In open foundations, with form work, at any level. Cement concrete for plain/reinforced concrete in open foundations as per drawing and technical specification. [MoRTH Specification : Section 1700 & 1500]. With 20mm nominal size aggregate R.C.C. M-25 grade	Abutment Piers For Wall 2 2 1	8.50 5.90 60.00	8.80 13.10 3.20	0.15 0.15 0.15	= = = Total = 22.44 cum 23.19 cum 28.80 cum 74.43 cum
6	Providing and laying steel reinforcement B-2 at any level In foundation complete as per drawing and technical specification. [MoRTH Specification : Section 1600]. HYSD bar reinforcement	2 2 2 1	8.20 4.75 5.60 60.00	8.50 8.50 12.80 3.10	0.70 0.80 1.40 0.55	= = = = Total = 97.58 cum 84.60 cum 200.70 cum 102.30 cum 465.18 cum
7	Structural cement concrete (Design Mix) B-2 In open foundations, with form work, at any level. Cement concrete for plain/reinforced concrete in open foundations as per drawing and technical specification. [MoRTH Specification : Section 1700 & 1500]. With 20mm nominal size aggregate P.C.C. M-15 grade	Abutment Piers Ded. Pier Shaft Ded. Abut. Shaft Ded. Pier Shaft Circular Por. 2 2 2 2 -2 -2 -2	33.40 8.20 36.80 5.60 8.50 4.75 0.79	0.15 8.50 0.15 12.80 0.80 8.50 0.64	0.70 0.80 1.40 0.60 0.60 0.60 0.60	= = = = = = = Total = 7.01 cum 111.52 cum 15.46 cum 86.02 cum -8.16 cum -48.45 cum -0.60 cum 162.79 cum
8	Structural cement concrete for plain concrete/ reinforced concrete for substructure at any level, complete as per drawing and technical specification. Design mix concrete -- [MoRTH Specification : Section 1500 & 1700]. PCC M-25 grade Upto 5 mtr. Height	Abt A1 A2 Paten wall Abt Cap A1 A2 1 1 2 2 2	8.50 8.50 4.20 4.20 8.50 3.50	1.05 1.05 0.75 0.75 0.90 0.70	4.20 4.20 6.00 6.15 0.30 0.70	= = = = = = Total = 37.49 cum 37.49 cum 37.80 cum 38.75 cum 4.59 cum 3.43 cum

S.No.	Description of Item.	No.	L	W	H/D	Quantity
		2	3.50	0.25	0.25	= 0.44 cum
		2	8.02	0.25	0.25	= 1.00 cum
		2	7.89	0.30	0.86	= 4.07 cum
		2	8.50	0.20	0.18	= 0.61 cum
		2	3.50	0.60	2.13	= 7.46 cum
		1	8.50	0.80	4.20	= 28.56 cum
		1	8.50	0.80	4.20	= 28.56 cum
		2	0.79	0.64	4.20	= 4.22 cum
	For Wall	2	8.50	1.00	0.30	= 5.10 cum
		1	60.00	0.45	4.20	= 113.40 cum
					Total	= 352.95 cum
9	Steel reinforcement in substructure					
B-3	complete. [MoRTH Specification : Section					
9.9.1	1600]					
	HYSD reinforcement					
	For Wall	0.012	352.95	7850.00		33248.21 Kg
	For Wall	0.01	119.35	7850.00		9368.98 Kg
	For Wall	0.01	126.00	7850.00		9891.00 Kg
					Total	= 52508.19 Kg
					Or	52.51 MT
10	Providing A.C. pipe Weep holes 150mm					
B-3	dia in Masonry/plain Concrete/Reinforced					
13	Concrete abutment, wing wall/return wall					
	complete as per drawing and technical					
	specification. [MoRTH Specification :					
	Clause 2205].					
		2	1	60.00		= 120.00 Mtr
		4	1	30.00		= 120.00 Mtr
	For Wall	4	0.45	100.00		= 180.00 Mtr
					Total	= 420.00 Mtr.
11	Providing and filling in foundation					
B-3	trenches, and back filling behind					
17.17	abutments, wing walls, return wall etc.,					
	and below pipe bed in layers not					
	exceeding 150mm thick including					
	watering and compacting (IRC-78)					
	complete as per drawing and technical					
	specifications (MoRTH specification 710					
	& 2200) including all lead and lifts.					
	Selected granular Material in filling					
	(MoRTH specification : Clause 2200).					
		2	7.90	2.30	2.00	= 72.68 cum
		2	4.20	2.30	2.00	= 38.64 cum
	For Wall	1	60.00	1.50	1.50	= 135.00 cum
					Total	= 246.32 cum
12	Filter medium behind abutment, wing wall					
B-1	and return wall, complete as per drawing					
4	and technical specification. (Same as					
	R7.20.3) [MoRTH Specification : Clause					
	2504].					
		2	7.95	0.60	2.00	= 19.08 cum
		2	4.20	0.60	2.00	= 10.08 cum
	For Wall	1	60.00	0.60	1.50	= 54.00 cum
					Total	= 83.16 cum
12	Structural RCC/PSC upto a height of 5					
B-4	mtr. for reinforced concrete in					
1.1.3.	superstructure complete including steel					
1	formwork, scaffolding etc. As per drawing					
	and technical specification. Design mix					
	concrete. [MoRTH Specification : 1500 &					
	1700]. RCC M-30 grade Solid slab					
	superstructure					
		3	10.00	8.50	0.75	= 191.25 cum

S.No.	Description of Item.	No.	L	W	H/D	Quantity
		3	10.00	0.79	0.56	= 13.19 cum
		2 3	8.50	0.70	0.20	= 7.14 cum
					Total	= 211.58 cum
13 B-4 2	Providing and laying HYSD bar reinforcement at any level in superstructure complete as per drawing and technical Specification. [MoRTH Specification : Section 1600]	0.012	211.58	7850.00		19930.65 Kg
					Or	19.93 MT
14 B-4 4	Reinforced cement concrete wearing coat M-30 grade at any level including formwork and reinforcement complete as per drawing and technical Specification. [MoRTH Specification : Clause 2204, 1500 & 1700].	1	40.00	8.50	0.08	= 25.50 cum
					Total	= 25.50 cum
15 B-4 15	Providing and Laying PCC M-15 grade levelling course below approach slab complete as per drawing and technical specifications. [MoRTH specification : 2700]	2	4.50	7.80	0.15	= 10.53 cum
					Total	= 10.53 cum
16 B-4 16	Reinforced cement concrete M-30 grade approach slab including reinforcement and formwork complete as per drawing and technical specifications. [MoRTH specification : 1500, 1600, 1700, 2704]	2	4.50	7.80	0.30	= 21.06 cum
					Total	= 21.06 cum
17 B-4 10.1	Providing, laying and fixing of Filler Joint Providing and fixing 2mm thick Corrugated Copper Plate in expansion joint complete as per drawing and technical specifications in MoRTH specification : clause 2605.	4	8.50	0.75		= 25.50 R.mtr
					Total	= 25.50 R.mtr
18 R-9 59	Supply of flood gauge (water gauge) as per IRC 67 standard 200cm long fixed on angle iron posts 50×50×6mm 2.6m long	2				= 2.00 Nos
					Total	= 2.00 Nos

S.No.	Description of Item.	No.	L	W	H/D	Quantity
19 R-7 2.1	Providing and Laying un-reinforced, dowel jointed, plain cement concrete pavement over a prepared sub-base with 43 grade cement, coarse and fine aggregate conforming to IS:383, maximum size of coarse aggregate not exceeding 25mm, mixed in a batching and mixing plant as per approved mix design, transported to site, laid with a fixed form or slip form paver, spread, compacted and finished in a continuous operation including provision of contraction, expansion, construction and longitudinal joints, joint filler, separation membrane, sealant primer, joint sealant, debonding strip, dowel bar, tie rod, admixtures and vacuum processing pump as approved, curing compound, finishing to lines and grades as per drawing as per MoRTH Specification : Clause No. 602, 112. Using Cement concrete M-20 pavement Grade					
		1.02	60.00			
				3.75	0.15	= 34.43 cum
				Total	=	34.43 cum

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